

GAYATRI KANCHERLA RAMESH

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Summary

Full-stack and Software Engineer experienced in building scalable web applications and data systems for enterprise environments. Delivered significant improvements in system speed and cost efficiency through successful migration and automation projects, and integrated AI features into real-time workflows. Skilled in Python, SQL, cloud platforms, and both frontend and backend development. Ready to leverage technical expertise to build scalable software systems and full-stack applications.

EDUCATION

California State University – Sacramento, CA

Aug 2023 - Dec 2025

MASTER OF SCIENCE, COMPUTER SCIENCE (GPA: 3.51/4)

- **Coursework:** AI/ML, Mobile and Cloud Pragmatics, Advanced DBMS, Computer System Structure, Cryptography, Object Oriented Programming System.

Amrita Vishwa Vidyapeetham University – Bengaluru, India

Jul 2018 - May 2022

BACHELOR OF TECHNOLOGY, COMPUTER SCIENCE (GPA: 3.48/4)

- **Coursework:** Operating Systems, Software Engineering, DBMS, Compiler Design, Java, Data Structures & Algorithms, Probability Theory, NLP, Deep Learning, ML and Data Mining, Big Data Analytics, Information Security.

SKILLS

- **Programming & Software Engineering:** C, C++, Java, Python, JavaScript, SQL, PHP, OOP, DSA
- **Full Stack Development:** MERN (MongoDB, Express.js, React, Node.js)
- **Frontend & UI Development:** ReactJS, AngularJS, HTML5, CSS3, Bootstrap
- **Backend, Cloud & DevOps:** Node.js, REST APIs, Express.js, CI/CD, Spring Boot, AWS, Azure, OCI, Docker, Kubernetes
- **AI/ML:** Machine Learning, Deep Learning, LLMs, Transformers, LSTM, NLP, Computer Vision, GenAI, Pandas, NumPy, TensorFlow
- **Data Analytics/Tools & Platforms:** Power BI, Data Analysis, Data Extraction, Data Validation, Statistical Analysis, ETL, Git, Microsoft Office (Excel, Word, PowerPoint), ServiceNow
- **Certifications:** IBM Data Science Professional Certificate, AWS Cloud Essentials, Microsoft Azure AI Fundamentals, Microsoft Azure Data Fundamentals, Codecademy Python 3 Course, HackerRank JavaScript Intermediate, SQL Advanced, Java, CSS, Python Basic Certificates

EMPLOYMENT HISTORY

California State University, ECS Dean's Office

Jan 2024 - Dec 2025

ECS IT CONSULTING STUDENT ASSISTANT

Sacramento, CA

- Diagnosed and resolved 100+ weekly cross-platform technical issues (Windows, Linux, macOS) using ServiceNow, improving system stability and reducing downtime for 1,000+ users.
- Deployed and maintained 40+ lab systems using SCCM, improving software reliability and lab uptime.
- Optimized ServiceNow workflows and documentation, reducing repeat issue frequency and improving ticket resolution times.

Foremost IT Solutions Private Limited

Feb 2025 - Oct 2025

AI SOFTWARE ENGINEER INTERN

Bengaluru, India

- Developed **AI-powered prototypes using Python and OpenAI APIs** to automate text processing and intelligent workflows for enterprise software applications.
- Built and integrated **REST APIs and backend services** to connect AI models with web applications, enabling real-time automation and data processing.
- Improved model performance through **data preprocessing, prompt optimization, and iterative testing**, increasing system responsiveness and prototype reliability.
- Collaborated with development teams to **design and test AI-enabled software features**, integrating machine learning models into web applications to automate business workflows and improve decision-support capabilities.

Trellix

Jan 2022 - Feb 2023

ASSOCIATE SOFTWARE DEVELOPMENT ENGINEER

Bengaluru, India

- Contributed to migrating the **historical event store from Elasticsearch to Snowflake**, building a scalable cross-product data warehouse and optimizing high-volume SQL queries, reducing latency by **45% (2.2s → 1.2s)**.
- Developed and refactored **frontend API integrations and backend data workflows**, improving platform responsiveness and contributing to **1.5× overall system performance and ~35% infrastructure cost reduction**.
- Diagnosed and resolved **enterprise customer escalations and critical software defects** by reproducing issues in Linux lab environments, analyzing logs, and collaborating with engineering teams through **JIRA** to deliver stable releases.
- Strengthened **security and reliability of Trellix cybersecurity products** by identifying and fixing **XSS vulnerabilities** while supporting enterprise security platforms (NX, EX, FX, AX, CM, PX-IA, Helix) used for threat detection and incident response.

Ramsai Food Products

Jan 2020 - Jan 2022

JUNIOR DATA SOFTWARE ENGINEER

Chittoor, India

- Cleaned and organized **production, inventory, and export datasets** using **SQL, Excel, and Power BI**, improving reporting accuracy and enabling faster decision-making for operations teams.
- Built **Power BI dashboards and automated reports** to monitor production output, shipment status, and inventory trends.
- Collaborated with operations and logistics teams to **analyze sales and export data**, identifying trends that improved demand forecasting and reduced reporting turnaround time.
- Implemented **data validation workflows and reporting pipelines**, improving data quality and increasing visibility for inventory planning and business decision-making.

ACADEMIC PROJECTS

AI-POWERED PERSONALIZED NUTRITION PLANNING SYSTEM

Jan 2025 - Nov 2025

- Built a multi-branch fusion model integrating health metrics, mental-wellbeing scores, genomic SNP sequences, and protein-folding data using LSTM + Transformer architectures to predict personalized nutrient requirements.
- Merged and preprocessed 73,778 dietary records from NHANES and 4,983 genomic profiles from the 1000 Genomes Project, handling missing values, scaling, data alignment, and multi-format feature construction.
- Designed an optimized multi-output regression system predicting grams, energy, protein, carbs, sugars, fiber, fat, and cholesterol with strong model convergence, statistical validation, and deployment-ready architecture.

VIRTUAL TRY ON

Jan 2022 - Apr 2022

- Developed a virtual fitting tool using OpenCV and HaarCascade, enhancing the online shopping experience by enabling realistic previews of clothing items.
- Improved customer engagement by 30% through increased confidence in purchase decisions.
- Implemented a feedback mechanism that collected user preferences, leading to targeted marketing strategies and a 20% increase in conversion rates.

PUBLICATIONS

- AIR WRITING RECOGNITION USING MEDIAPIPE AND OPENCV. *Conference Paper, Jul 2022; International Conference on Ubiquitous Computing and Intelligent Information Systems; Link: https://link.springer.com/chapter/10.1007/978-981-19-2541-2_35*